

ELK 346 E Data Acquisition & Embedded Systems

2023 – 2024 Spring Semester
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Topics

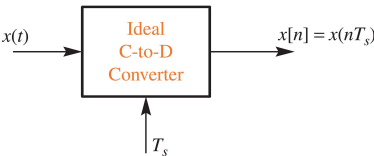
- Sensors
- Filters, ADCs, DACs
- Sampling
- Fundamental Digital Electronics
- Basics of Microcontrollers – Registers, ALU
- ARM - Introduction
- ARM – Registers
- Embedded Programming

Sampling

- Sampling breaks the continuously varying analogue input signal or signals into a **finite number of data points** which can be processed by a computer.
- Information about the signal in between the points is **lost forever**.
- As more data points necessitate more memory space and faster acquisition, this is yet another example of the **trade-off between quality and cost**.
- The signal is not only sampled in time, but its **voltage is quantized** into a finite number of levels.
- The system designer must choose the **optimum sized quanta** for each.

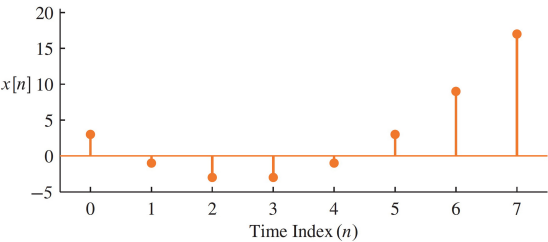
Sampling

Block diagram representation of the ideal continuous-to-discrete (C-to-D) converter. The parameter T_s specifies uniform sampling of the input signal every T_s seconds.



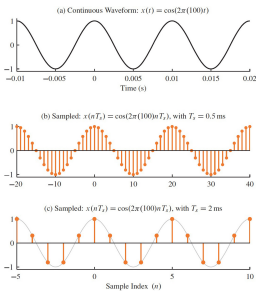
Sampling

Plotting format for discrete-time signals, called a stem plot. In MATLAB, the function stem produces this plot.



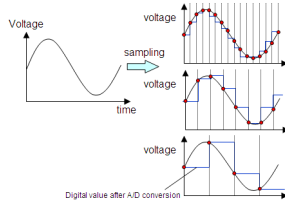
Sampling

A continuous-time 100 Hz sinusoid (a) and two discrete-time sinusoids formed by sampling at $f_s = 2000$ samples/s (b) and at $f_s = 500$ samples/s (c).



Important Terms

- **Conversion Time** : The time it takes for the ADC to complete the conversion of an analog signal to a digital output value.
- **Sampling Frequency** : The rate at which the analog input signal is sampled and converted into a corresponding digital output value per second.
- **Resolution** : The number of bits used to represent the digital output value, which determines the smallest increment of analog input that can be distinguished from noise or other errors.
- **Nyquist Rate** : the minimum sampling rate required to accurately represent a signal, which is twice the highest frequency component of the signal according to the Nyquist-Shannon sampling theorem.

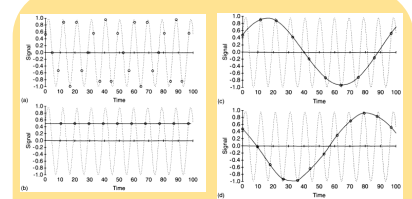


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Aliasing

- Aliasing in sampling occurs when a continuous signal is **sampled at a rate that is too low to capture its high-frequency content**.
- As a result, high-frequency components of the signal are **incorrectly represented** by lower frequency components in the sampled signal, leading to distortions and errors in the reconstructed signal.



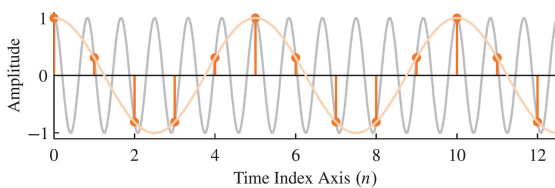
Aliasing in the time domain: (a) $N > 2$, correct sampling; (b) $N = 1$, sampling frequency eliminated; (c) $N < 1$, aliased signal; (d) $-N < -2$, negative aliased frequency, (.....) original signal; (—) aliased signal.

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Aliasing

Illustration of aliasing: two continuous-time signals drawn through the same samples. The samples belong to two different cosine signals with different frequencies, but the cosine functions have the same values at $n = 0, 1, 2, 3, \dots$. Aliasing occurs when a high frequency is indistinguishable from a low frequency after sampling.

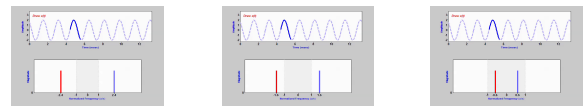


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Aliasing and Folding

- This illustrates the phenomenon of **aliasing**. A 600 Hz sinusoid is sampled at 500 samples per second.
- This illustrates the phenomenon of **folding**. A 600 Hz sinusoid is sampled at 750 samples per second.
- This illustrates that the Sampling Theorem is true. In this case, a 600 Hz sinusoid is sampled at 2000 samples per second. Since the samples are taken at more than two times the frequency of the cosine wave, there is no aliasing.

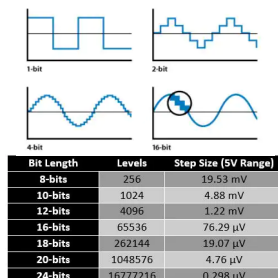


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Resolution

- The decision of choosing what resolution ADC depends on the **accuracy requirements of the application** and the available system resources such as power, speed, and cost, with **higher resolution generally offering greater accuracy but requiring more resources**. Factors such as the input signal bandwidth, noise, and dynamic range should also be considered to determine the appropriate resolution for the specific application.



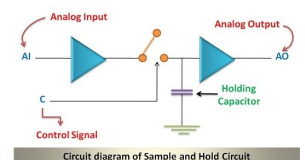
$$\text{ADC Voltage Resolution} = \frac{(V_H - V_L)}{2^n}$$

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Sample and Hold

- A sample and hold circuit is an electronic circuit that samples an input signal and then holds the sampled value constant for a period of time.
- It is typically used in analog-to-digital converters (ADCs) to acquire and hold a voltage signal before it is converted to a digital value.

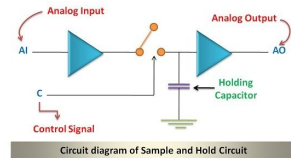


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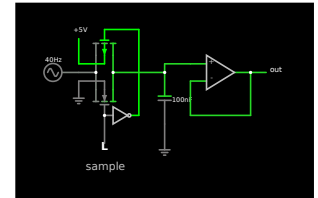
Sample and Hold

- The basic operation of a sample and hold circuit involves two stages: the **sample stage** and the **hold stage**.
- In the **sample stage**, an analog input signal is sampled and held by a capacitor. The capacitor is charged to the input voltage level during the sampling phase.
- In the **hold stage**, the capacitor is disconnected from the input source and connected to a buffer amplifier, which maintains the voltage level across the capacitor constant.



Sample and Hold

- Simulation
- <https://www.falstad.com/circuit/e-samplenhold.html>



Practical Example

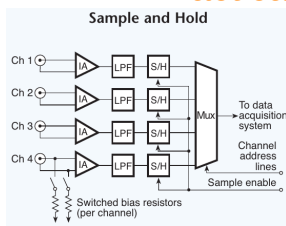
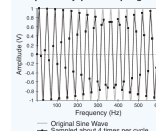


Fig. 3.05. A four-channel multiplexer with low-pass filters and simultaneous sample-and-hold circuits ensures that all the samples are taken within a few ns of each other.

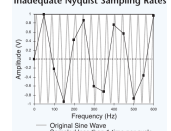
- Simultaneous sample and hold circuit (SS&H)
- Each input signal passes through an instrumentation amplifier (IA), a low-pass filter, and into a sample-and-hold buffer (S/H).
- When the sample enable line goes high, each S/H samples its input signal and holds it while the multiplexer switches through the readings.

Nyquist Theorem

Acceptable Nyquist Sampling Rates



Inadequate Nyquist Sampling Rates



- The Nyquist Sampling Theorem states that a signal can be completely reconstructed from its samples if the signal is sampled at a rate that is **at least twice the highest frequency present in the signal**.
- This means that the minimum sampling rate required to accurately capture a signal is known as the Nyquist rate, and it is equal to twice the highest frequency component of the signal.
- For example, if a signal has a highest frequency of 10 kHz, it must be sampled at a rate of at least 20 kHz to avoid losing any information. If the sampling rate is too low, some of the signal's frequency components may not be captured, resulting in distortion or loss of the original signal.
- ADC Sampling Frequency has to be chosen according to the maximum frequency component of the signal to be sampled.

Sampling Freq. Example

- What should be the minimum sampling frequency for the signal below?

$$x(t) = \cos(10\pi t) + 4\cos(30\pi t)$$

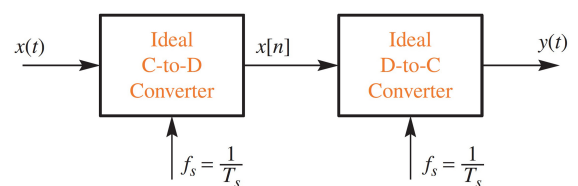
- There are 2 frequency component in this signal

$$\begin{aligned} \omega_1 &= 10\pi \text{ rad/s} & f_1 &= 5 \text{ Hz} \\ \omega_2 &= 30\pi \text{ rad/s} & f_2 &= 15 \text{ Hz} \end{aligned}$$

- So the min. sampling frequency should be 30 Hz

Reconstruction

Sampling and reconstruction system where the rates of the C-to-D and D-to-C converters are the same.



Over-Sampling / Under-Sampling

- Under-sampling**, also known as aliasing, is the process of sampling a signal at a rate **lower than its Nyquist rate**, which is the minimum sampling rate required to accurately represent the signal. This can result in loss of information or distortion of the original signal.
- Over-sampling** is the process of sampling a signal at a rate **higher than its Nyquist rate**. This can improve the accuracy of the signal representation and reduce the noise in the signal. Over-sampling is often used in signal processing applications such as digital signal processing, data acquisition, and control systems.
- In summary, over-sampling increases the number of samples per unit time, while under-sampling decreases the number of samples per unit time.

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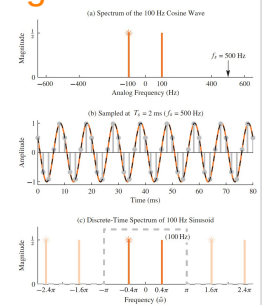
Over-Sampling

Over-sampling a 100 Hz sinusoid at $f_s = 500$ samples/s.

(a) Continuous-time spectrum;

(b) time-domain plot, showing the samples $x[n]$ as gray dots, the original signal $x(t)$ as a continuous **orange** line, and the reconstructed signal $y(t)$ as a dashed black line, which is the 100 Hz sinusoid passing through the same sample points;

(c) discrete-time spectrum plot, showing the positive and negative frequency components of the original sinusoid at $\omega = \pm 0.4\pi$ rad, along with two aliases.



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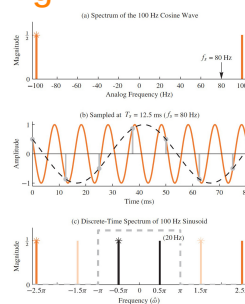
Under-Sampling 1

Under-sampling a 100 Hz sinusoid at $f_s = 80$ samples/s.

(a) Continuous-time spectrum;

(b) time-domain plot, showing the samples $x[n]$ as gray dots, the original signal $x(t)$ as a continuous **orange** line, and the reconstructed signal $y(t)$ as a dashed black line, which is a 20 Hz sinusoid passing through the same sample points;

(c) discrete-time spectrum plot, showing the positive and negative frequency components of the original sinusoid at $\omega = \pm 2.5\pi$ rad, along with two sets of alias components.



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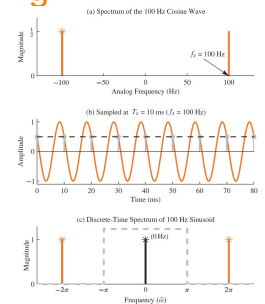
Under-Sampling 2

Under-sampling a 100 Hz sinusoid at $f_s = 100$ samples/s.

(a) Continuous-time spectrum;

(b) time-domain plot, showing the samples $x[n]$ as gray dots, the original signal $x(t)$ as a continuous **orange** line, and the reconstructed signal $y(t)$ as a dashed black line, which is a 0 Hz sinusoid passing through the same sample points;

(c) discrete-time spectrum plot, showing the positive and negative frequency components of the original sinusoid at $\omega = \pm 2\pi$ rad, along with parts of two sets of alias components that are superimposed at $\omega = 0$ to give a DC value of $\frac{1}{2}e^{j\pi/3} + \frac{1}{2}e^{-j\pi/3} = \cos(\pi/3)$.



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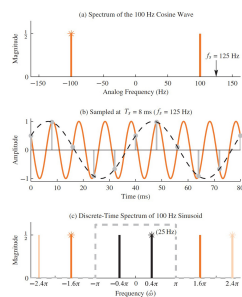
Under-Sampling 3

Under-sampling a 100 Hz sinusoid at $f_s = 125$ samples/s.

(a) Continuous-time spectrum;

(b) time-domain plot, showing the samples $x[n]$ as gray dots, the original signal $x(t)$ as a continuous **orange** line, and the reconstructed signal $y(t)$ as a dashed black line, which is a 25 Hz sinusoid passing through the same sample points;

(c) discrete-time spectrum plot, showing the positive and negative frequency components of the original sinusoid at $\omega = \pm 1.6\pi$ rad, along with two sets of aliases. Note how the phase is negated in this example of aliasing by folding.



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Matlab Example

```
Fs = 1000; % Sampling frequency
t = 0:1/Fs:1-1/Fs; % Time vector
x = sin(2*pi*50*t);

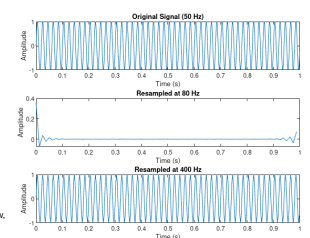
% Plot the original and resampled signals
figure;
subplot(3,1,1); plot(t, x);
title('Original Signal (50 Hz)');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,1,2);
Fs_new = 80; plot(0:1/Fs_new-1/Fs_new, x_resampled_80);
title('Resampled at 80 Hz');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,1,3);
Fs_new = 400; plot(0:1/Fs_new-1/Fs_new, x_resampled_400);
title('Resampled at 400 Hz');
xlabel('Time (s)'); ylabel('Amplitude');

% Resample the signal at 80 Hz
Fs_new = 80;
x_resampled_80 = resample(x, Fs_new, Fs);

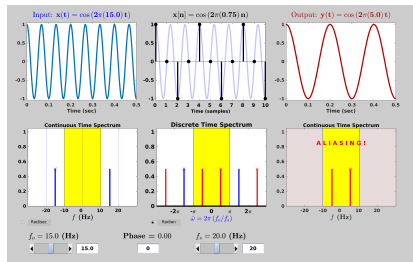
% Resample the signal at 400 Hz
Fs_new = 400;
x_resampled_400 = resample(x, Fs_new, Fs);
```



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Matlab Demo



<https://dspfirst.gatech.edu/matlab/#con2dis>

Research

- What are the ADC/DAC capabilities of STM32 F1, F4 and F7 series?
- What are their resolutions?
- What's the sampling frequency range?
- How to adjust sampling frequency?
- What is prescaler?